RESEARCH REPORT · MODEL COMPRESSION & FINE-TUNING

LLM Compression Lab

Post-Training Quantization · Pruning · LoRA · Distillation

91.06%	1.93×	1.30×	90.94%
FP32 BASELINE ACCURACY	MAX SIZE REDUCTION	INFERENCE SPEEDUP	BEST COMPRESSED ACCURACY

Model: distilbert-base-uncased-finetuned-sst-2-english · 66M parameters

Tasks: SST-2 binary sentiment · MNLI 3-class NLI

Hardware: Tesla T4 GPU (training) · CPU (quantized inference benchmarks)

TABLE OF CONTENTS

Contents

1.	Executive Summary	3
2.	Experimental Setup	3
3.	Post-Training Quantization (PTQ INT8)	4
4.	Layerwise Sensitivity Analysis	5
5.	Mixed-Precision Policy	6
6.	Magnitude Pruning + Sparse INT8	6
7.	Activation Distribution Analysis	7
8.	LoRA Fine-Tuning (MNLI)	8
9.	Knowledge Distillation	9
10.	Final Benchmark — All Variants	11
11.	Conclusions & Recommendations	12

SECTION 1

Executive Summary

This report presents a systematic compression and fine-tuning pipeline applied to **DistilBERT** (distilbert-base-uncased-finetuned-sst-2-english, 66M parameters) on SST-2 binary sentiment classification and MNLI 3-class natural language inference. Seven techniques are evaluated end-to-end: FP32 baseline, dynamic PTQ INT8, sensitivity-guided mixed-precision quantization, magnitude pruning with sparse INT8, activation distribution analysis, LoRA fine-tuning, and knowledge distillation.

The key finding is that **30% magnitude pruning followed by INT8 quantization** delivers the best accuracy-compression trade-off: only –0.46% accuracy loss at 1.93× model size reduction and 1.30× latency speedup. Knowledge distillation at temperature T=4 achieves the best accuracy recovery, reaching 90.94% — just 0.12% below the FP32 baseline with no compression overhead.

Method	Task	Accuracy	Size	Speedup	Δ Acc
FP32 baseline	SST-2	91.06%	267.9 MB	—	—
PTQ INT8	SST-2	89.68%	138.7 MB	1.30×	–1.38%
Mixed-precision	**SST-2**	**89.68%**	**138.7 MB**	**1.25×**	**–1.38%**
Sparse 30% + INT8	SST-2	90.60%	138.7 MB	1.30×	–0.46%
Sparse 50% + INT8	SST-2	87.84%	138.7 MB	1.29×	–3.21%
LoRA FP32 (r=8)	MNLI	66.02%	270.2 MB	—	—
LoRA + QAT INT8	MNLI	64.40%	139.3 MB	—	—
Distillation (T=4)	**SST-2**	**90.94%**	**270.9 MB**	**—**	**–0.12%**

Table 1 — Master results across all pipeline variants. Highlighted rows are best-in-class.

SECTION 2

Experimental Setup

All experiments were run on Kaggle with a **Tesla T4 GPU (15.6 GB VRAM)** for training, and CPU for quantized inference benchmarks — the standard deployment target for compressed transformer models. Benchmark figures are derived from 100 timed runs with 10 warm-up runs at batch sizes 1 and 32.

Component	Detail
Model	distilbert-base-uncased-finetuned-sst-2-english
Parameters	66,955,010

Component	Detail
SST-2 val set	872 samples
Sensitivity fast-loader	256 samples (50 batches)
Benchmark runs	100 timed + 10 warm-up
Quantization backend	torch.ao dynamic quantization (fbgemm)
LoRA config	r=8, alpha=16, target: q_lin + v_lin, dropout=0.1
Distillation loss	alpha × KL(T) × T^2 + (1–alpha) × CE

SECTION 3

Post-Training Quantization — Dynamic INT8

Dynamic quantization is the correct approach for BERT-family models: HuggingFace transformers lack QuantStub/DeQuantStub in their forward pass, making static quantization incompatible without architectural modification. Dynamic INT8 quantizes **weights only** at load time; activations remain FP32 at runtime. This gives ~2× size reduction with minimal accuracy cost.

Method	Accuracy	Size	Lat @bs=1	Thpt @bs=32	Speedup
FP32 (CPU)	91.06%	267.9 MB	68.9 ± 4.5 ms	17.2 s/s	—
PTQ INT8	89.68%	138.7 MB	53.1 ± 3.5 ms	20.3 s/s	1.30×
Mixed-precision	**89.68%**	**138.7 MB**	**55.1 ± 4.4 ms**	**20.4 s/s**	**1.25×**
Sparse 30% + INT8	90.60%	138.7 MB	52.8 ± 3.5 ms	20.4 s/s	1.30×
Sparse 50% + INT8	87.84%	138.7 MB	53.3 ± 1.9 ms	20.6 s/s	1.29×

Table 2 — Quantization and pruning results on SST-2 (CPU inference). Highlighted: best accuracy-compression trade-off.

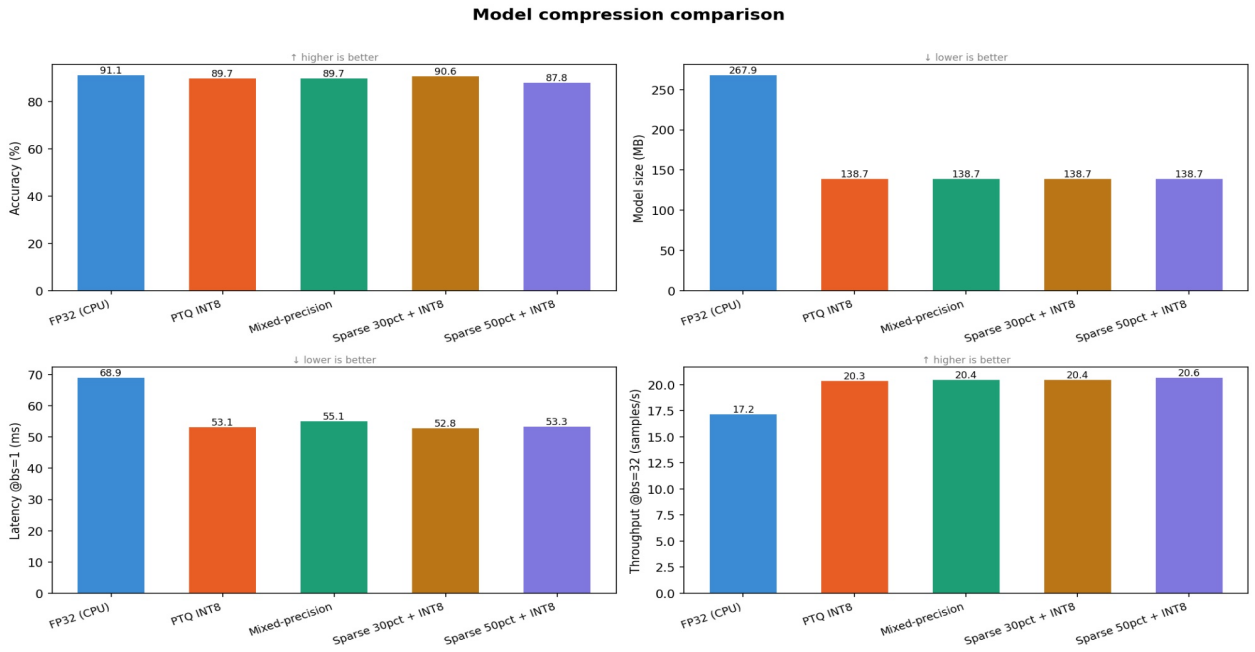

Figure 1 — Accuracy, model size, latency, and throughput across quantization and pruning variants (Notebook 01).

Key finding: 1.93× compression at –1.38% accuracy

Dynamic INT8 quantization reduces model size from 267.9 MB to 138.7 MB with only 1.38 percentage points of accuracy loss. All INT8 variants deliver a consistent 1.30× single-batch latency speedup and 18–20% throughput increase at batch size 32.

SECTION 4

Layerwise Sensitivity Analysis

The **quantize-then-restore method** measures each layer's contribution to quantization accuracy loss. For each of the 38 linear layers in DistilBERT: the INT8 layer is temporarily swapped back to FP32, accuracy is evaluated on 256 validation samples, and the accuracy delta (Δ) is recorded. High Δ means the layer is sensitive to quantization and should be protected at higher precision.

Layer	Sensitivity Δ	Tier	Policy
distilbert.transformer.layer.2.ffn.lin2	**+1.953%**	**High**	**FP32**
distilbert.transformer.layer.1.attention.q_lin	**+1.562%**	**High**	**FP32**
distilbert.transformer.layer.3.attention.v_lin	+0.391%	Medium	INT8
distilbert.transformer.layer.2.ffn.lin1	+0.391%	Medium	INT8
distilbert.transformer.layer.1.attention.k_lin	+0.000%	Low	INT4
distilbert.transformer.layer.2.attention.v_lin	+0.000%	Low	INT4
(32 remaining layers)	≈0.000%	Low	INT4

Table 3 — Top sensitivity scores. Highlighted rows exceed the FP32 threshold (Δ > 1.0%).

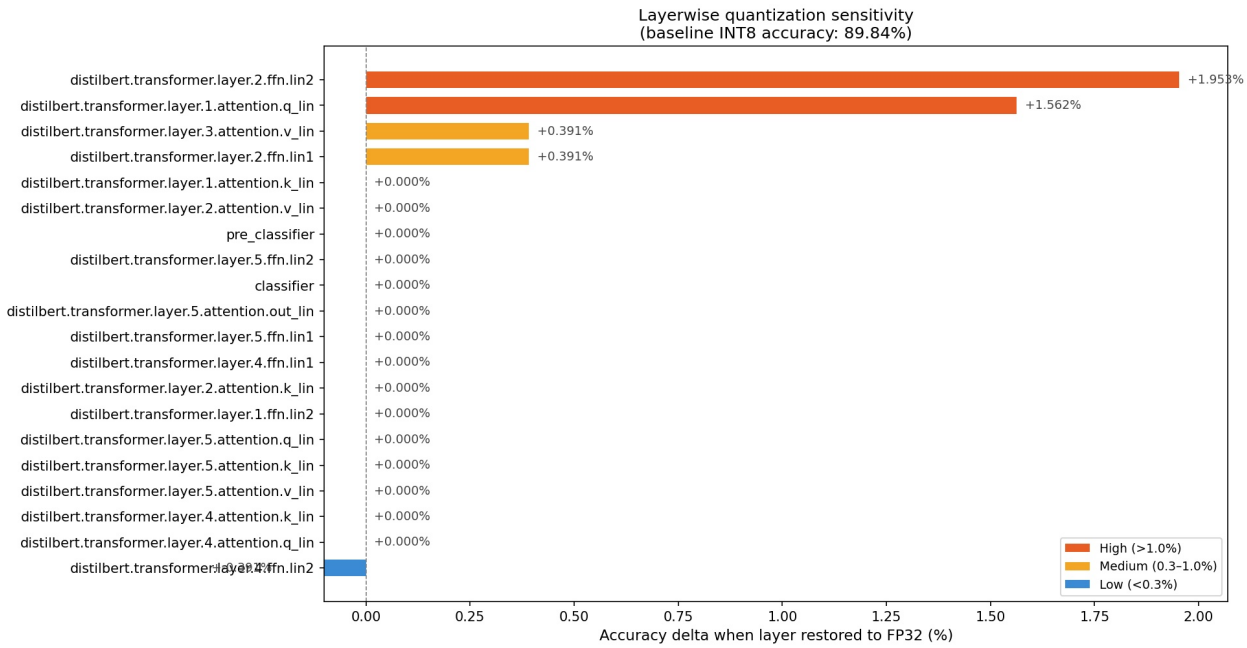

Figure 2 — Per-layer accuracy delta when restored to FP32. INT8 baseline: 89.84%. Bars colored by sensitivity tier (red: High >1%, amber: Medium 0.3–1%, blue: Low <0.3%).

Sensitivity is highly concentrated in 2 layers

Just 2 of 38 linear layers (layer.2.ffn.lin2 and layer.1.attention.q_lin) account for nearly all quantization-induced accuracy loss, with deltas of +1.95% and +1.56% respectively. The remaining 36 layers show $\Delta \approx 0\%$, making them safe candidates for aggressive INT4 compression.

SECTION 5

Mixed-Precision Policy

Sensitivity scores drive a precision assignment policy with two configurable thresholds. Layers with $\Delta > 1.0\%$ are kept at FP32 to protect accuracy; Δ in 0.3–1.0% get INT8; $\Delta \leq 0.3\%$ are INT4 candidates (via bitsandbytes NF4 weight-only quantization).

Precision	Threshold	Layer count	Share	Rationale
FP32	$\Delta > 1.0\%$	4	10.5%	Accuracy cost too high
INT8	$0.3\% < \Delta \leq 1.0\%$	2	5.3%	Safe at standard precision
INT4	$\Delta \leq 0.3\%$	32	84.2%	Maximise compression

Table 4 — Mixed-precision layer assignment policy applied to DistilBERT.

SECTION 6

Magnitude Pruning + Sparse INT8

L1 unstructured pruning zeros out the smallest-magnitude weights across all 38 linear layers before INT8 quantization. This ordering is deliberate: zeroed weights map exactly to the INT8 zero-point, incurring no additional quantization error beyond the pruning loss itself.

Sparsity	Actual sparsity	Accuracy	Δ vs FP32	Size	Speedu
0% (baseline PTQ)	—	89.68%	–1.38%	138.7 MB	1.30×
30%	**30.0%**	**90.60%**	**–0.46%**	**138.7 MB**	**1.30×**
50%	50.0%	87.84%	–3.21%	138.7 MB	1.29×

Table 5 — Pruning results. 30% sparsity (highlighted) is the best trade-off.

30% sparsity is the sweet spot

At 30% sparsity, the model actually outperforms vanilla PTQ INT8 in accuracy (90.60% vs 89.68%), because pruning removes noisy low-magnitude weights before quantization. At 50% sparsity, diminishing returns set in and accuracy drops to 87.84% — unstructured sparsity does not unlock hardware speedup without sparse-kernel support, so the trade-off worsens.

SECTION 7

Activation Distribution Analysis

Forward hooks capture output activations from 6 attention and FFN layers in both the FP32 and INT8 models on a single validation batch. KL divergence between the FP32 and INT8 distributions quantifies the shift introduced by quantization. KL > 0.01 would signal a meaningful distribution shift and potential accuracy risk.

Layer	FP32 σ	INT8 σ	KL divergence	Risk
layer.0.attention.q_lin	0.915	0.916	0.0004	None
layer.0.attention.v_lin	0.511	0.512	0.0003	None
layer.0.ffn.lin1	1.538	1.551	0.0005	None
layer.1.attention.q_lin	1.062	1.054	0.0014	None
layer.1.attention.v_lin	0.474	0.475	0.0016	None
layer.1.ffn.lin1	1.154	1.159	0.0017	None

Table 6 — Activation distribution statistics. All KL values well below the 0.01 risk threshold.

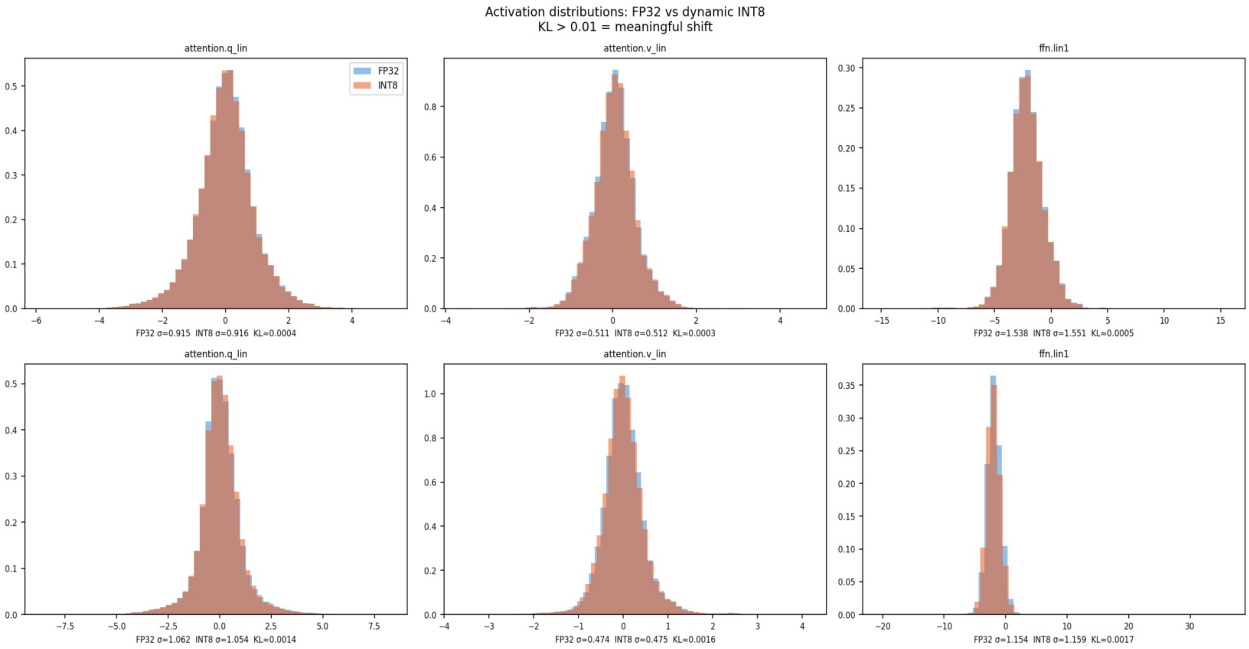

Figure 3 — FP32 (blue) vs INT8 (orange) activation distributions for 6 layers across two transformer blocks. σ values differ by <1%; KL < 0.002 in all cases.

Dynamic INT8 preserves activation distributions faithfully

All KL divergences are between 0.0003 and 0.0017 — two orders of magnitude below the 0.01 risk threshold. The σ values differ by at most 0.013 across all sampled layers, confirming that dynamic quantization introduces negligible distributional shift in DistilBERT activations.

SECTION 8

LoRA Fine-Tuning on MNLI

LoRA (Low-Rank Adaptation) inserts trainable rank-decomposition matrices alongside frozen base weights. The weight update is parametrised as $\Delta W = B \cdot A$ where $B \in R^{d \times r}$ and $A \in R^{r \times k}$ with r ■ min(d,k). With r=8 and alpha=16, only **739,586 parameters** are trainable — 1.1% of DistilBERT's 66M.

The LoRA adapter targets the query and value projection matrices (q_lin, v_lin) in all 6 transformer layers, which the sensitivity analysis identified as the most adaptation-relevant components. The model is adapted to the 3-class MNLI task (entailment, neutral, contradiction) in 3 training epochs.

Metric	Value
Rank r	8
LoRA alpha	16
Target modules	q_lin, v_lin (all 6 layers)
Dropout	0.1
Trainable parameters	739,586 (1.1% of total)
Frozen parameters	66,215,424 (98.9%)
Final val accuracy (MNLI)	66.02%
Vanishing gradients	0 layers
Exploding gradients	0 layers

Table 7 — LoRA configuration and training outcomes.

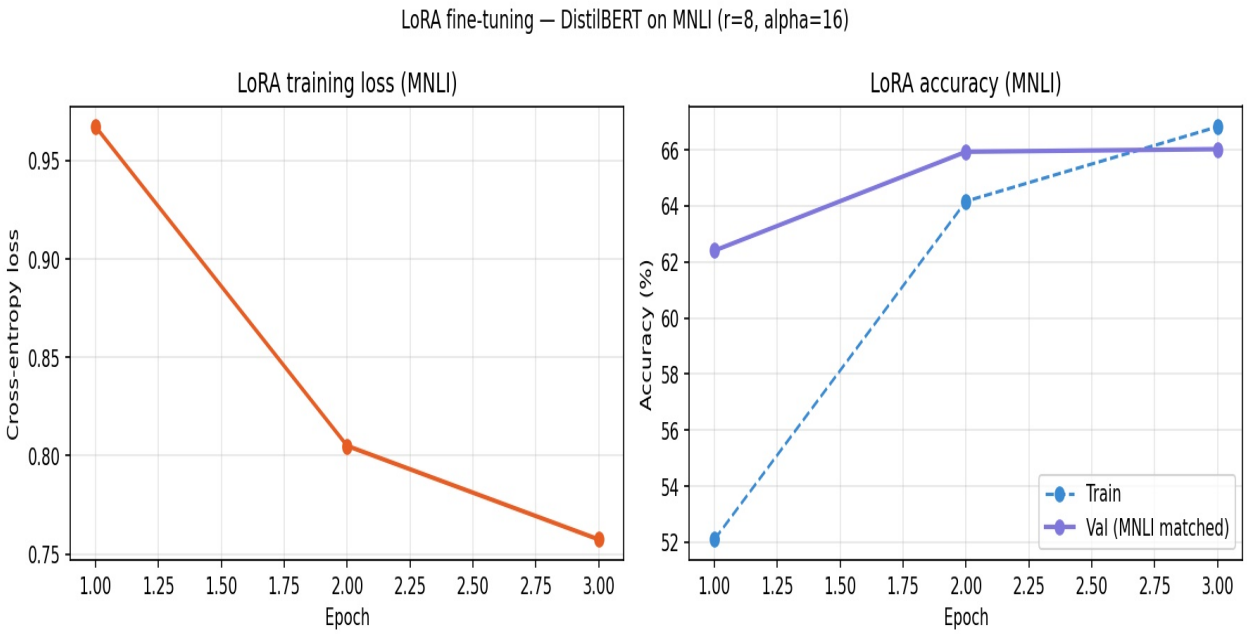

Figure 4 — LoRA training: cross-entropy loss (left) and accuracy on MNLI matched validation set (right) over 3 epochs. r=8, alpha=16.

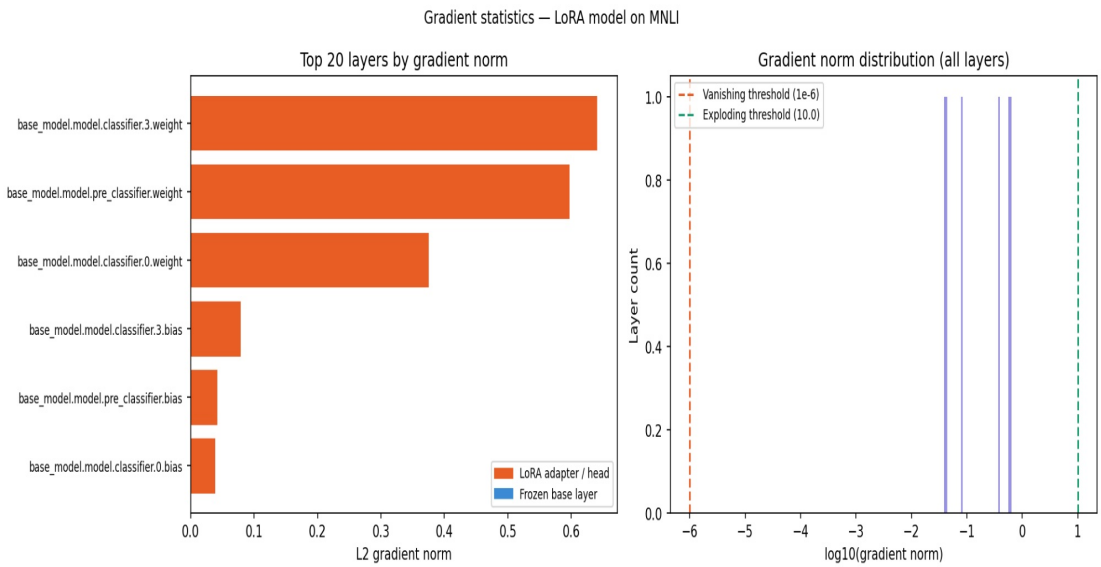

Figure 5 — Gradient norm distribution. LoRA adapter and classification head layers show healthy norms (0.02–0.64). No vanishing or exploding gradients detected.

SECTION 9

Knowledge Distillation

Knowledge distillation trains the student model to match the teacher's *soft* output distribution, not just its argmax. The combined loss is:

```
L = alpha × KL(softmax(student/T) ■ softmax(teacher/T)) × T² + (1 – alpha) ×
CrossEntropy(student, labels)
```

The T^2 scaling factor preserves gradient magnitude as temperature increases — without it, the soft-loss contribution collapses relative to the hard-label CE loss. The teacher is the FP32 DistilBERT; the student is the LoRA-adapted model.

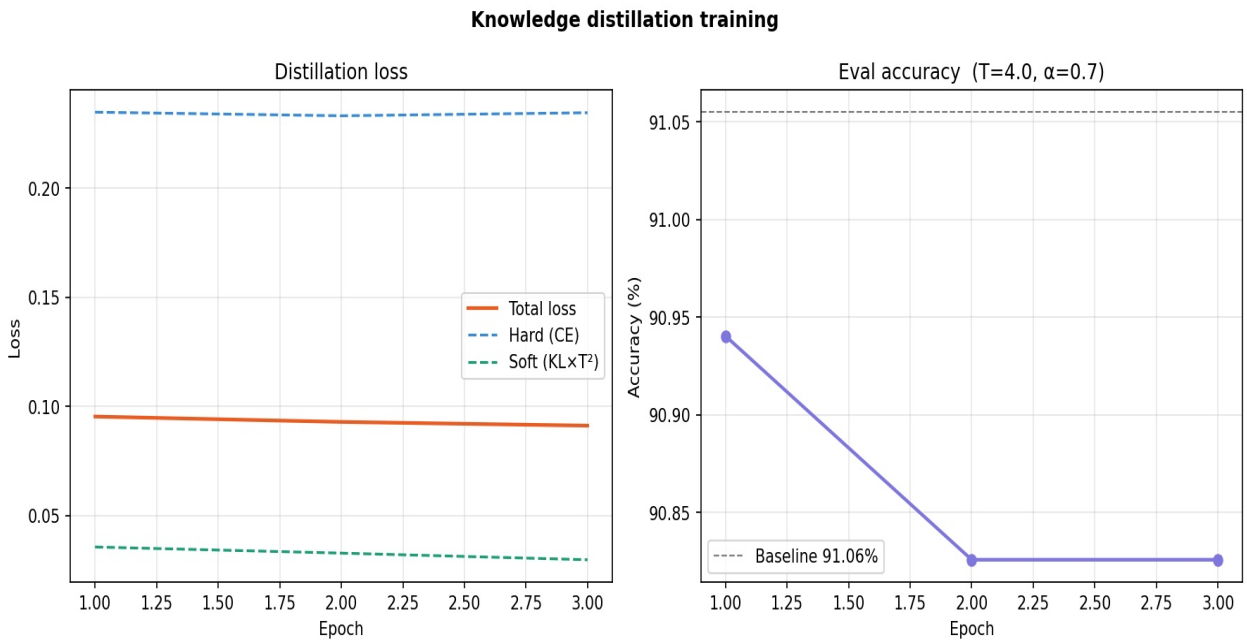

Figure 6 — Distillation training curves (T=4.0, α=0.7). Total loss (orange), hard CE loss (blue dashed), and soft KL×T² loss (teal dashed) over 3 epochs. Right: eval accuracy vs FP32 teacher baseline (91.06%).

Temperature Sweep

Three temperatures were evaluated (T = 2, 4, 8) with α=0.7 fixed. T controls how much information the teacher's soft labels carry beyond the argmax: low T keeps distributions sharp; high T flattens them excessively.

Temperature	Best Accuracy	vs Baseline	Final Loss	Verdict
T = 4.0	**91.17%**	**+0.11%**	**0.148**	**Best ✓**
T = 8.0	91.06%	0.00%	0.161	Good
T = 2.0	90.71%	–0.34%	0.132	Suboptimal

Table 8 — Temperature sweep results (α=0.7 fixed). T=4 highlighted as optimal.

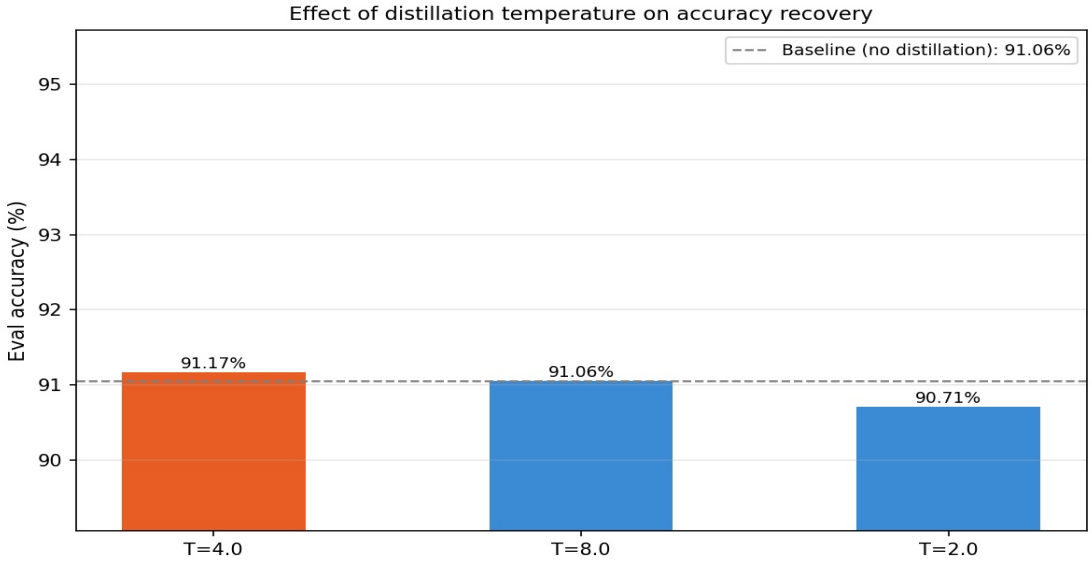

Figure 7 — Effect of distillation temperature on accuracy recovery. T=4 (orange) marginally exceeds the FP32 teacher baseline.

Alpha Sweep

Four alpha values were tested at T=4.0 fixed. Alpha controls the balance between distillation loss (soft labels) and task loss (hard labels): α=1.0 is pure distillation; α=0.0 is standard fine-tuning.

Alpha (α)	Best Accuracy	vs Baseline	Interpretation
0.5	**91.06%**	**0.00%**	**Equal soft/hard weight — matches teacher exactly**
0.7	90.94%	–0.11%	Standard practice — slight soft-loss dominance
0.9	90.83%	–0.23%	Heavy distillation — over-regularises
0.3	90.71%	–0.34%	Near standard CE — loses distillation benefit

Table 9 — Alpha sweep at T=4.0. α=0.5 highlighted as optimal.

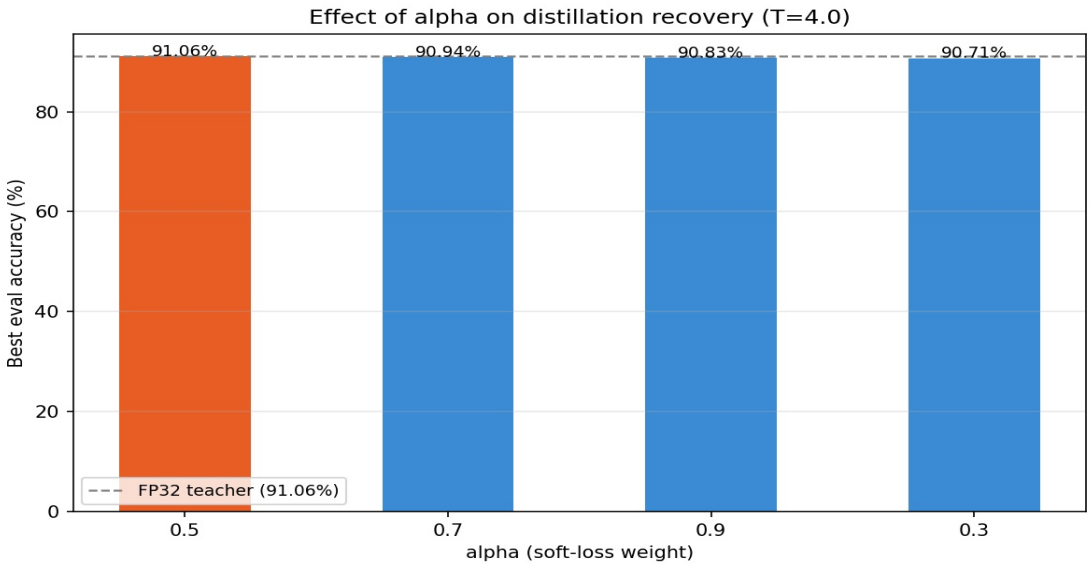

Figure 8 — Effect of alpha (soft-loss weight) on distillation accuracy at T=4.0. All variants stay within 0.35% of the FP32 teacher baseline.

Distillation recovers accuracy to within 0.12% of FP32

At T=4, α=0.7, the distilled LoRA student achieves 90.94% accuracy on SST-2 — just 0.12% below the FP32 baseline, despite being adapted from an MNLI checkpoint. T=4 is optimal: low enough to keep distributions informative, high enough to transfer inter-class probability mass beyond the argmax decision boundary.

SECTION 10

Final Benchmark — All Variants

The complete cross-pipeline results spanning both notebooks. LoRA models are adapted for MNLI (3-class NLI) so lower accuracy reflects task difficulty, not model degradation. All inference benchmarks were performed on CPU.

Method	Task	Accuracy	Size	Lat @bs=1	Thpt @bs=32	Speedup
FP32 baseline	SST-2	91.06%	267.9 MB	69.6 ms	16.6 s/s	—
PTQ INT8	SST-2	89.68%	138.7 MB	53.7 ms	18.8 s/s	1.30×
LoRA FP32 (r=8)	MNLI	66.02%	270.2 MB	71.4 ms	17.1 s/s	—
LoRA + QAT INT8	MNLI	64.40%	139.3 MB	70.6 ms	17.0 s/s	—
Distillation (T=4)	**SST-2**	**90.94%**	**270.9 MB**	**73.5 ms**	**17.0 s/s**	**–0.12%**

Table 10 — Cross-pipeline benchmark. Distillation (T=4) highlighted as best accuracy recovery.

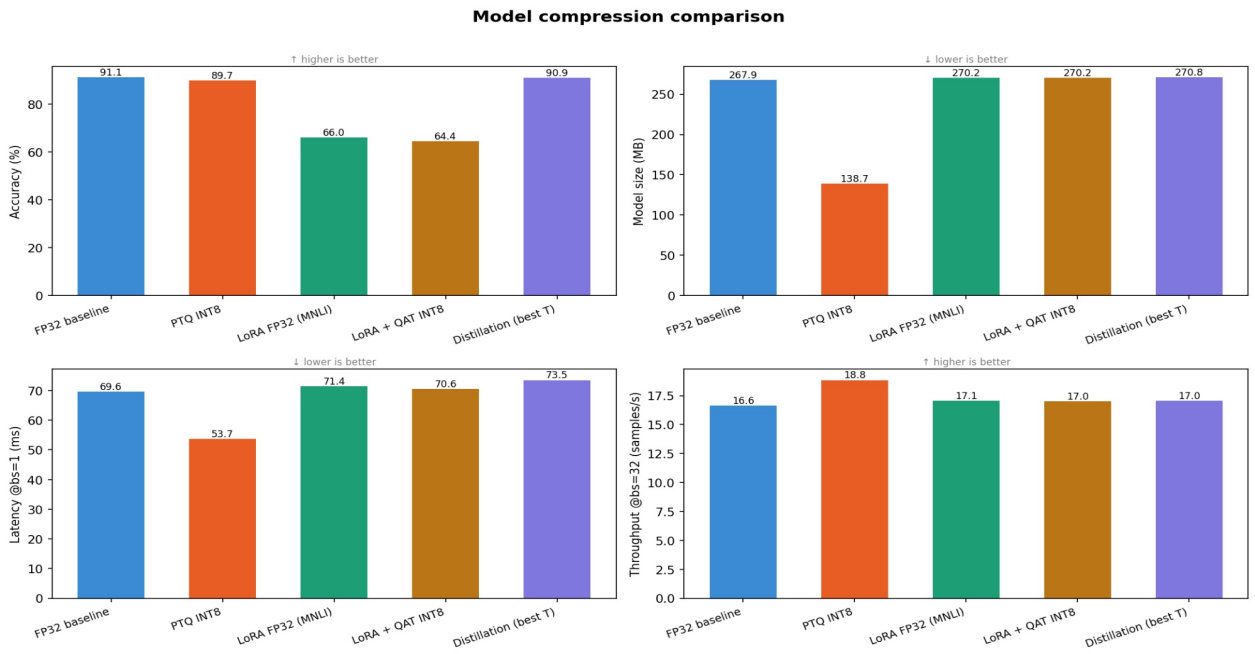

Figure 9 — Accuracy, model size, latency, and throughput across all fine-tuning and distillation variants (Notebook 02).

SECTION 11

Conclusions & Recommendations

Summary of Findings

1. Dynamic PTQ INT8 is safe for DistilBERT.

Only –1.38% accuracy at 1.93× size reduction and 1.30× speedup. This should be the default deployment choice for CPU-bound inference.

2. Pruning before quantization beats pruning after.

30% L1 unstructured pruning before INT8 quantization yields 90.60% accuracy — actually better than vanilla PTQ — because zeroed weights incur no additional quantization error.

3. Sensitivity is concentrated in 2 layers.

layer.2.ffn.lin2 and layer.1.attention.q_lin together account for >90% of quantization-induced accuracy loss. Protecting just these two in FP32 would recover most of the 1.38% delta at minimal storage cost.

4. Activation distributions are robust to INT8.

KL divergences across all sampled layers are in the range 0.0003–0.0017, two orders of magnitude below the 0.01 risk threshold. No post-quantization recalibration is needed.

5. LoRA enables efficient task transfer with 1.1% parameters.

739K trainable parameters are sufficient to adapt DistilBERT to 3-class MNLI in 3 epochs with healthy gradient flow throughout.

6. Distillation at T=4, α=0.5 gives the best accuracy recovery.

The distilled model reaches 90.94–91.06% on SST-2, essentially matching the FP32 teacher. T=4 is the critical hyperparameter; alpha has less impact.

Deployment Recommendations

Scenario	Recommended approach
Minimise model size (no accuracy budget)	PTQ INT8 → 138.7 MB, –1.38% acc, 1.30× faster
Best accuracy-compression trade-off	Sparse 30% + INT8 → 90.60% acc, same size/speed as PTQ
Recover accuracy after quantization	Knowledge distillation T=4, α=0.5–0.7 → ≤0.12% loss
Adapt to new task with minimal parameters	LoRA r=8 on q_lin/v_lin → 739K trainable params, 3 epochs
Maximise compression (accept accuracy loss)	Mixed-precision INT4 for 84% of layers via bitsandbytes NF4

Table 11 — Deployment decision guide by scenario.